Abstention as a First-Class Output: Measuring the Silent Errors an Evidence Pipeline Would Otherwise Ship

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Abstract

Pipelines that assemble public biomedical records into scientific claims are evaluated on what they emit, not on what they decline. We show that this misses the dominant failure mode. A *silent error* is an output that is wrong, well-formed, raises no exception, and is indistinguishable downstream from a correct result. We introduce *silent-error ablation*, a protocol that removes a pipeline's verification guards and counts what the unguarded pipeline then emits over identical inputs. We apply it to a rare-disease drug repurposing pipeline in which every stage must verify its input or decline it, and declining is a counted output. Over 746 clinical variant records for 8 genes, we find that removing residue verification makes 16 of 114 emitted protein sequences silently wrong (14.0%). Removing copy-number exclusion misattributes 382 of 746 records to genes they are not evidence about, moving 3 of 8 genes from apparent support to zero. We report the cost as well: the sequence lane declines 266 of 364 records. We also report two defects the protocol found in our own instrumentation, including a rate we had overstated by ten percentage points. Neither was visible without a mechanical check.

1 Introduction

Computational drug repurposing produces rankings. Recent systems score millions of drug–disease pairs [Huang et al., 2024], and commercial databases sell curated competitive landscapes. These systems are evaluated on the quality of what they emit. What they generally do not produce is a record of which evidence produced a score, or a principled way to report that the evidence was insufficient; a confident guess and an auditable finding are rendered identically.

This matters because the dominant failure mode in an evidence-assembling pipeline is not a crash or a mis-ranking. It is a *silent error*: an output that is (i) wrong, (ii) syntactically well-formed, (iii) raises no exception or warning, and (iv) cannot be distinguished by a downstream consumer from a correct output. Silent errors are the non-statistical analogue of hallucination—plausible, confident, and undetectable at the point of use. They are also, by construction, invisible to any evaluation that scores only the outputs a pipeline chooses to produce.

We take a narrow position. We do not claim to rank better. We claim that a pipeline can be built so that each stage must either *verify* its input against a stated precondition or *decline* it, with declining recorded as a first-class, counted output, and that this discipline prevents a measurable class of silent error. Our contributions:

1. **Silent-error ablation**, an evaluation protocol for evidence pipelines. Rather than scoring the pipeline's recommendations—which here would require wet-lab validation we have not

performed—we score its *refusals*, by removing each verification guard and counting what the resulting pipeline emits over identical inputs.

2. A **verify-or-abstain architecture** in which the decision logic is a small set of readable rules, no model output may set a verdict, and every record is content-addressed under a single Merkle root that a reader re-derives client-side.

3. **Measurements** on 746 real public records, reported with their cost and their limits, including two defects the protocol exposed in our own instrumentation.

2 Related work

Repurposing at scale. Graph-based models predict drug–disease associations across large biomedical knowledge graphs [Huang et al., 2024]. These target coverage and accuracy; provenance and calibrated refusal are not their objective, and the outputs are not designed to be independently re-derived.

Selective prediction. The option to abstain has a long history in classification, from the reject-option formulation [Chow, 1970] to selective classification with coverage guarantees [El-Yaniv and Wiener, 2010, Geifman and El-Yaniv, 2017]. That literature abstains on the basis of model confidence. Our abstentions are not confidence-based: they are triggered by a failed precondition on the *input record*, which is checkable, deterministic, and reportable with a reason.

Provenance and tamper-evidence. We adopt the Merkle audit-tree construction of Certificate Transparency [Laurie et al., 2013], which gives inclusion proofs and makes any post-hoc edit detectable.

Missing data. Our own defect is best described in Rubin's taxonomy [Rubin, 1976]: the loss was *missing not at random*, since the probability of a record being dropped was a function of its own size, which correlated with its class.

3 System

Input and output. The input is a rare-disease name. The output is a set of (disease, target, drug-programme) rows, each carrying a verdict in {GAP, NOT_A_GAP, ABSTAIN}, the identifier of the rule that produced it, and a receipt: the content digests of every record the verdict consumed.

Evidence acquisition. Five public sources are queried, one question each: disease biology and target tractability; clinical variant records; registered trials, retaining status and phase so that a terminated trial is not read as an untried opportunity; regulatory approval; and cell-morphology profiles. Every response is hashed at capture and its query URL recorded, so a third party can re-issue the query and compare. A source whose responses cannot be re-obtained by a third party is inadmissible; one candidate source is excluded on exactly this ground and is reported as excluded rather than quietly omitted.

Identity normalisation. Symbols resolve by exact match or an explicit alias table; there is no fuzzy matching. This matters because collisions are real: one symbol in our corpus denotes two unrelated proteins, and a similarity match would return a valid gene, succeed at every downstream step, and answer a question about the wrong molecule.

Decision. Nine first-match rules assign the verdict. No model output may set a verdict, resolve a symbol, or choose a threshold; a language model is used only to phrase already-decided rows.

Custody. Every record is content-addressed and committed under one RFC 6962 Merkle root [Laurie et al., 2013]. Verification re-derives the root in the reader's browser from the record bytes; a single-byte edit moves the root and the first divergent node is named. This is a property of the construction, not an experimental result, and we report it as such.

3.1 The verification guards

Two guards carry the measurements in this paper.

G1, residue verification. A variant record names a wild-type residue and a position. Before applying the substitution to a canonical protein sequence, the pipeline checks that the named residue is what the canonical sequence actually has at that position. If not, the record is numbered against a different transcript, and the row abstains.

G2, copy-number exclusion. A variant record is admitted as evidence about a gene only if it is gene-specific. Records typed as copy-number events, or spanning more than a small number of genes, are excluded: a deletion covering hundreds of genes is not evidence about any one of them.

4 Evaluation

4.1 Setup and metric

We compare two pipelines over identical inputs: **enforcement on**, the system as described, and **enforcement off**, the same pipeline with one guard removed. The off condition is not a strawman: it is the implementation obtained by following each source's documentation without anticipating the specific failure, and no guard other than the ablated one is disabled.

We report the *silent error rate*: the fraction of outputs the unguarded pipeline emits that are demonstrably wrong. Failures that are *loud*—a crash, an out-of-range index—are excluded from the numerator, because a loud failure is not the problem being studied. Absolute counts accompany every rate; the denominators are small and hiding that would misrepresent them.

Corpus: pathogenic and likely-pathogenic clinical variant records for 8 genes, captured 2026-08-25, subset digest `3aa5e6723563`. Acquisition reconciles: 746 records fetched = 364 admitted + 382 excluded + 0 unavailable.

4.2 Results

Table 1: Silent errors prevented. Rates are over what the unguarded pipeline would emit; loud failures are excluded from the numerator.

Guard	Naive emits	Silently wrong	Rate	Enforced emits
G1 residue verification	114	16	14.0%	98
G2 copy-number exclusion	746	382	51.2%	364

Table 1 gives the headline result. Without G1, 16 of 114 emitted sequences are wrong. Each is valid FASTA built from real residues. A structure-prediction tool accepts it without complaint and returns a structure for a protein that does not exist. With G1, 98 sequences are emitted (40 substitutions, 58 truncations) and all 98 were verified position-by-position against their wild-type reference.

Without G2, 382 of 746 records are credited to genes they are not evidence about. The per-gene effect (Table 2) is not uniform, and this is the more interesting result: 3 of 8 genes (PGS1, PHB2, CHCHD3) move from apparent evidentiary support to an honest zero. An unguarded pipeline reports them as supported, and the top-ranked "evidence" for one of them is a chromosomal deletion syndrome.

Table 2: Per-gene effect of G2. * genes reduced to zero.

Gene	Unfiltered	After G2	Excluded
TAFAZZIN	339	109	230
CRLS1	33	3	30
PGS1*	14	0	14
PTPMT1	13	1	12
HADHA	267	251	16
PHB	0	0	0
PHB2*	46	0	46
CHCHD3*	34	0	34

4.3 The cost of abstention

A trivial policy—decline everything—also achieves a zero silent error rate and is useless. The guards must therefore be shown to decline *selectively.* The sequence lane declines 266 of 364 records (Table 3); most declines are records carrying no protein-level change or frameshifts that cannot be reconstructed from the record at all, neither of which any substituting pipeline could emit. Of the records that are genuinely substitution-eligible, the guard declines 16 and emits 98, all verified correct.

Table 3: Why the sequence lane declines. Only *residue mismatch* reflects a record a substituting pipeline would otherwise have emitted.

Reason declined	Records
frameshift	131
no protein notation	118
residue mismatch	16
position out of range	1

5 What the audit found

We applied the protocol to our own instrumentation before submission. Two findings bear directly on the results above. We report them because omitting them would be an instance of the failure the paper describes.

The pipeline was discarding records without counting them. A reconciliation check—fetched records must equal admitted plus excluded plus unavailable—revealed that 100 records had been dropped by a single unlogged branch. The cause was a hard upstream limit: the source refuses to serialise a response above a fixed size, and one record listing over a thousand genes pushed its batch past that limit. Our error handler read three possible error keys; the source reports this one under a fourth, so our own log recorded only a generic "no result" string. *We had built a guard that detected the loss and destroyed the explanation.* The loss was *missing not at random* [Rubin, 1976]—the records lost were lost because they were large, and they were large because they spanned many genes, which is precisely the class the guard exists to exclude. Left uncorrected, it biased our own headline G2 rate downward by roughly six percentage points.

Our first reported rate was wrong by ten points. An independent recount disagreed with the pipeline's own count of G1 rejections. The pipeline checked the residue precondition *before* classifying the variant, so frameshift records that also mismatched were filed as residue mismatches—inflating the reported figure with records that no substituting pipeline could emit. The published rate was 24.6%; the correct rate is 14.0%. The emitted sequences were byte-identical before and after the fix: the guard had rejected exactly the same records, and only the reasons were wrong. This was a *measurement* defect rather than a behavioural one, which is why nothing appeared broken and why only recounting exposed it.

We draw a narrow conclusion from both. Mechanical reconciliation—inputs equal outputs plus rejects, asserted at every boundary—caught defects that careful reading did not, including in the instrument that produces this paper's numbers.

6 Limitations

The G1 effect is one gene. All 16 residue mismatches occur in a single gene; another gene contributing 58+ emitted sequences produces zero. The measured G1 result therefore reflects one transcript-numbering convention, not 8 independent observations. $n = 1$, not $n = 8$.

The rejected records are recoverable, and we discard them. Testing the failing records rather than the passing ones showed that a single constant offset of +30 residues reconciles all 16 exactly: they are correctly numbered against a longer transcript. They are silently wrong only when applied to the canonical sequence. Abstention is the safe response here, not the best one; a pipeline that detected the offset would recover them.

Small denominators and a single corpus. Rates are computed over hundreds, not thousands, of records, from one snapshot of one source. The source is live: two honest runs a day apart returned different totals, which is why we pin a capture date and publish the digest of the exact table used.

We evaluate our own guards on data we selected. The guards were designed by us, and the failures they catch are failures we anticipated. This measures the value of enforcement given a correct precondition; it does not measure how many preconditions we failed to think of. The audit above is evidence that this residual is not zero.

We do not evaluate the science. Whether any proposed pairing is biologically correct requires wet-lab validation not performed here. The system's claim ceiling is enforced in code and permits a repurposing hypothesis, never a therapeutic claim.

7 Conclusion

Provenance and abstention are usually argued for on principle. We argue for them with a number: on real public records, removing two verification guards causes a pipeline to emit 16 of 114 sequences that are silently wrong and to misattribute 382 of 746 variant records. The intervention is architectural rather than statistical—verify or abstain, enforced in code—and so transfers to any pipeline that assembles evidence into claims, whether or not a language model is in the loop. The most persuasive evidence we can offer is that the discipline caught our own instrument twice, and that neither failure was visible without it.

Future work: extending custody to the turn level of the model interactions that assist the pipeline, so that assistance itself carries receipts; and recovering rather than merely declining the transcript-offset records.

Reproducibility statement

Every figure in this paper is generated from the pipeline outputs by a script; no number is typed by hand. A second script refuses to let a superseded figure survive in the prose. An anonymised artifact containing the code, the captured corpus with digests, and a one-command regeneration of every table is available to reviewers [link withheld for anonymous review].

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